

Stage 1 Tutorial Activity

- Why Software Engineering Still Matters

Stage 1 | Introducing Software Technology Case Study with Python and Guided AI use

Learning goals

- Explain why software engineering is broader than coding.
- Identify stakeholders in a simple software problem.
- Recognise missing requirements.
- Critically evaluate AI-generated feature suggestions.
- Explain why AI output should not automatically be treated as correct.

Activity 1 - Think-Pair-Share (10 minutes)

If ChatGPT or Copilot can produce a 100-line Python application very quickly, what knowledge does a software engineer still need?

1. Read, understand, and critically think/analyze Product/User requirements.
2. Design System Architecture, optimal methods to solve the system, and maintain the system.
3. Evaluate if AI's output is correct, review it, AI's direction, critique it, nudge it in the right direction, test it in real world and secure it accordingly.

Activity 2 - Is This Software Engineering? (10 minutes)

Scenario A: A student writes a 50-line Python calculator.

Scenario B: A team develops a payroll system used by 5,000 employees.

Scenario C: An AI assistant generates a simple appointment application from one prompt.

(Software Engineering for me means, not only programming, but optimally approaching the problem, designing the architecture, scaling the system, maintain and test the system etc. which usually happens in production based real world system, so used that as a reason below.)

Scenario	Programming?	Software engineering?	Why?
A	Yes	No (could be though)	It is not used in a real world production system by real users, if it is then it is Software Engineering.
B	Yes	Yes	It is a big system, used as a real world production system.
C	Yes	No.	Only generating doesn't mean it classifies as Software Engineering,

			especially, with just one prompt, there is no testing, securing, fool-proofing system, applying fallbacks, maintaining over time etc.
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Activity 3 - SmartCare Problem Analysis (20 minutes)

Client statement: SmartCare Community Clinic currently uses spreadsheets and paper records to manage patients and appointments. The clinic wants new software to improve these processes.

Task 1 - Identify stakeholders

Stakeholder	What do they need?
Patients	Easy appointment booking, accurate records, privacy, and no clashes.
GP/practitioners	Quick access to patient information and appointment schedules, again no clashes.
Receptionist	Efficient patient registration, searching, displaying, and appointment management (CRUD).
Clinic / IT Devs	Clinic should provide the requirements for what system they need exactly, devs need that PRD to build a correct system.

Task 2 - Identify current problems

1. Patient information is stored across spreadsheets and paper records, making it difficult to manage consistently.
2. Appointments are difficult to schedule (conflict free), update, and track.
3. No way to list all appointments, or search for a record, difficult for GP and receptionist to get the information fast.
4. Paper records increase the risk of errors, loss, or unauthorised access, and privacy is really important.

Task 3 - Ask client questions

1. What sort of data about patients needs to be stored?
2. Who should be allowed to view, edit, delete, create patient records?
3. How are appointments currently booked, resolved in case of conflict, and rescheduled on what basis?
4. Should patients be allowed to view available GP and timeslots?
5. How is the information provided on GP's end? Are they allowed to reschedule appointments too?

Activity 4 - Critique an AI Response (15 minutes)

An AI assistant suggests: appointment management; facial-recognition login; AI diagnosis recommendations; patient search; online payment; practitioner schedule view; insurance processing; automatic treatment-plan generation.

Suggestion	Client evidence?	In scope?	Decision
Appointment management	Yes	Yes	Yes
Facial recognition login	No	No	No, It is an extension/overkill.
AI diagnosis recommendations	No	No	No, It is an extension/assumption/o verkill.
Patient search	Yes	Yes	Yes
Online payment	No	No	No, It is an extension/assumption, but can include if patients are allowed to book and need payments in advance.
Practitioner schedule view	Yes (Probably)	Yes (Probably)	Yes
Insurance processing	No	No	No, It is an extension/assumption
Treatment-plan generation	No	No	No, It is an extension/assumption

Exit question

Write one activity that a software engineer must perform and that cannot safely be delegated entirely to AI.

A: One activity that we must perform is reviewing the system requirements with clients. AI can suggest requirements, but it cannot safely determine on its own what the clinic actually needs (or at least whats in scope), what is legally or ethically appropriate, or what risks the system may create. Also, we must review AI generated code since, specifically for healthcare systems, the data is extremely crucial to protect and privacy is super important, so system needs to be risk averse.